Chapter 8

Revision and consolidation

8.1 Revision strategy

Key points

Effective revision alternates retrieval from memory, explicit correction of mistakes and reuse in a problem. Passively rereading the course is not enough: you must attempt a question without a model, compare your answer, then redo the question after a delay.

For each chapter, use the following code: **V** (mastered), **O** (shaky), **R** (to redo). After the diagnosis, work first on an orange or red skill, then check immediately with a similar question.

Skill	V	O	R	Remedial action
Choosing a model	☐	☐	☐	Redo a sorting exercise: arithmetic/geometric/exponential.
Reading and calculating	☐	☐	☐	Redo the targeted fluency exercises.
Writing up and interpreting	☐	☐	☐	Add data, formula, calculation, unit and concluding sentence.

Session 52 — Diagnosis of what has been learned

Revision and consolidation

11 May 2027 • B: 2 h – C: 1 h

Session objectives

- Identify what has been learned.
- Classify the mistakes.
- Plan the remediation.

Fluency – 6 to 8 minutes, no calculator

1. Write the multiplier for an increase of 25 %. $1 + \dfrac{25}{100} = 1.25$
2. Give the formula for the general term of an arithmetic sequence starting from u_0. $u_n = u_0 + nr$
3. Solve $10^x = 1000$. $x = 3$
4. Differentiate $x + \dfrac{4}{x}$. $1 - \dfrac{4}{x^2}$
5. State the law of total probability with two branches. $P(B) = P(A)P_A(B) + P(\overline{A})P_{\overline{A}}(B)$

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: identify what has been learned and classify the mistakes.

Investigation

Answer 14 questions without the course, two per chapter. Record each mistake in a grid: knowledge, choice of method, calculation, reading, write-up. Then choose two personal priorities.

Detailed solution – teacher edition

The correction expects a profile of mistakes and a plan for catching up, not just an overall mark. Have students say aloud which piece of data triggers the choice of method.

Written record

The diagnosis is used to choose the remediation: recognise the mistake, recall the useful rule, redo a similar question, then check without help.

Worked example

Take the data from the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer is in the teacher edition.

Detailed solution – teacher edition

The correction expects a profile of mistakes and a plan for catching up, not just an overall mark.

Practice – essential

1. Choose between an arithmetic sequence, a geometric sequence and an exponential function in three situations.
2. Solve an exponential equation using a logarithm.
3. Calculate a probability from a tree, then interpret it.

Detailed solution – teacher edition

Justify each choice by absolute or relative change; check positivity before taking the logarithm; multiply along a path and conclude with a sentence.

Textbook references

Further reading: Reactivation of chapters 1 to 7. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 29, exercise 52

In France, since 2016, sales of hybrid vehicles have risen sharply. The table below gives the number of hybrid vehicles sold in France per year.

Year	2016	2017	2018
Number of hybrid vehicles sold	7 420	11 010	15 350

Part A

1. Determine the percentage change between 2016 and 2017, then between 2017 and 2018.
2. Determine the geometric mean of 1.48 and 1.39.
3. What is the average annual rate of change between 2016 and 2018?

Part B

Assume that from 2018 onwards, sales of hybrid cars increase by 43 % per year and that this trend will continue. The sequence (u_n) models the change in sales, where the integer part of u_n represents the number of hybrid cars sold during year $(2018 + n)$.

1. What is the value of u_0? Calculate u_1 and u_2.
2. What type of sequence is (u_n)? Express u_n in terms of n.
3. According to this model, how many hybrid vehicles will be sold in 2023?

Detailed solution – teacher edition

The rates are approximately 48.4 % and 39.4 %. The geometric mean of 1.48 and 1.39 is $\sqrt{1.48 \times 1.39} \approx 1.434$, that is an average rate of about 43.4 %. In part B, $u_0 = 15\,350$, $u_n = 15\,350 \times 1.43^n$; for 2023, $n = 5$ and the integer part of u_5 gives approximately 91 788 vehicles.

Homework

Redo practice exercise 3 with the following data: replace the initial value by 380 and the rate or common ratio by 6. Write it up in four lines: data, formula, calculation, conclusion.

Detailed solution – teacher edition

The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and give a confidence level from 1 to 4. Essential question(s) of the day: 1 and 2.

Method

1 h session: 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Session 53 — Reactivation: exponential functions

Revision and consolidation

13 May 2027 • B: 1 h – C: 2 h

Session objectives

- Reactivate exponential models.
- Solve and interpret.
- Check the units.

Fluency – 6 to 8 minutes, no calculator

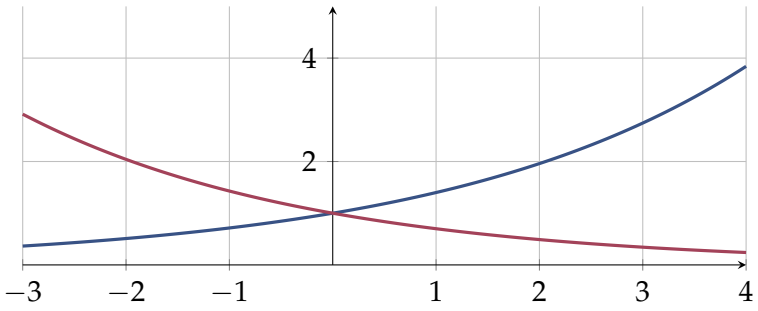
1. Simplify $3x + 7 - x + 2$. $2x + 9$
2. Solve $10^x = 1000$. $x = 3$
3. Differentiate $x + \dfrac{4}{x}$. $1 - \dfrac{4}{x^2}$
4. State the law of total probability with two branches. $P(B) = P(A)P_A(B) + P(\overline{A})P_{\overline{A}}(B)$
5. Give the formula for the general term of an arithmetic sequence starting from u_0. $u_n = u_0 + nr$

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: reactivate exponential models and solve and interpret.

Investigation

A value of 18 000 increases by 3.5 % per year. Write the model, calculate the value after eight years and determine when it will exceed 25 000.



Detailed solution – teacher edition

$V(t) = 18\,000 \times 1.035^t$, $V(8) \approx 23\,703$. The threshold is exceeded from the tenth year onwards. Have students say aloud which piece of data triggers the choice of method.

Written record

An exponential function models a constant relative change. First identify the initial value and the multiplier.

Worked example

Take the data from the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer is in the teacher edition.

Detailed solution – teacher edition

$V(t) = 18\,000 \times 1.035^t$, $V(8) \approx 23\,703$. The threshold is exceeded from the tenth year onwards.

Practice – essential

1. Choose between an arithmetic sequence, a geometric sequence and an exponential function in three situations.
2. Solve an exponential equation using a logarithm.
3. Calculate a probability from a tree, then interpret it.

Detailed solution – teacher edition

Justify each choice by absolute or relative change; check positivity before taking the logarithm; multiply along a path and conclude with a sentence.

Textbook references

Further reading: Calao, p. 48, no. 1–8, then p. 54, no. 47–50. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 54, exercise 58

The number of members of a basketball club increased over the last three years by 2.5 %, then by 4.1 %, and finally by 3.8 %. Calculate the average annual rate of change.

Detailed solution – teacher edition

The overall multiplier is $1.025 \times 1.041 \times 1.038$. The average annual multiplier is its cube root, approximately 1.0346. The average annual rate of change is therefore approximately 3.46 %.

Homework

Redo practice exercise 2 with the following data: replace the initial value by 385 and the rate or common ratio by 7. Write it up in four lines: data, formula, calculation, conclusion.

Detailed solution – teacher edition

The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and give a confidence level from 1 to 4. Essential question(s) of the day: 1, 2 and 3.

Method

1 h session: 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Session 54 — Reactivation: common logarithm (base 10)

Revision and consolidation

18 May 2027 • B: 2 h – C: 1 h

Session objectives

- Reactivate logarithms.
- Solve an equation.
- Check the conditions.

Fluency – 6 to 8 minutes, no calculator

1. Calculate 5 % of 160. 8
2. Differentiate $x + \dfrac{4}{x}$. $1 - \dfrac{4}{x^2}$
3. State the law of total probability with two branches. $P(B) = P(A)P_A(B) + P(\overline{A})P_{\overline{A}}(B)$
4. Give the formula for the general term of an arithmetic sequence starting from u_0. $u_n = u_0 + nr$
5. Solve $10^x = 1000$. $x = 3$

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: reactivate logarithms and solve an equation.

Investigation

Solve $10^x = 340$, $1.06^n = 2$ and $1.06^n \geq 2$. For each answer, distinguish the real value, the approximate value and the minimum integer index.

Detailed solution – teacher edition

$x \approx 2.5315$; $n \approx 11.90$; for the integer threshold, $n = 12$. Have students say aloud which piece of data triggers the choice of method.

Written record

The logarithm turns an exponential equation into an equation in the exponent. Conditions: base positive and different from 1, right-hand side positive.

Worked example

Take the data from the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer is in the teacher edition.

Detailed solution – teacher edition

$x \approx 2.5315$; $n \approx 11.90$; for the integer threshold, $n = 12$.

Practice – essential

1. Choose between an arithmetic sequence, a geometric sequence and an exponential function in three situations.
2. Solve an exponential equation using a logarithm.
3. Calculate a probability from a tree, then interpret it.

Detailed solution – teacher edition

Justify each choice by absolute or relative change; check positivity before taking the logarithm; multiply along a path and conclude with a sentence.

Textbook references

Further reading: Calao, p. 70, no. 1–10, then pp. 73–75, no. 30–42. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 76, exercise 43

Let x be a strictly positive real number. The aim of the exercise is to determine an order of magnitude of x.

1. $x = 23.54$ and $y = 569.47$.
   a. The order of magnitude of a number is the power of 10 closest to that number. Give the order of magnitude of x and y.
   b. Using the calculator, determine $\log(x)$. Conjecture a link between $\log(x)$ and the order of magnitude of x. Check your conjecture with y.
2. $x = 5.4517$. Using the calculator, determine the order of magnitude of x and $\log(x)$. Is the conjecture made in question 1 true?
3. Suppose that x is written in scientific notation: $x = a \times 10^n$ where $a \in [1;10[$ and n is an integer. Give the order of magnitude of x depending on the value of a.
4. Deduce a method linking the order of magnitude of a strictly positive real number with its common logarithm (base 10).
5. Deduce the order of magnitude of the following two numbers: 2.26^{102} and 365.256^{2019}.

Detailed solution – teacher edition

23.54 is closer to 10 than to 100, whereas 569.47 is closer to 1 000 than to 100; their orders of magnitude are 10 and 10^3. The change occurs at $a = \sqrt{10}$, not at $a = 5$. In scientific notation $a \times 10^n$, the order of magnitude is 10^n if $a < \sqrt{10}$ and 10^{n+1} if $a \geqslant \sqrt{10}$. We apply this rule to the fractional part of the logarithms: $102\log(2.26) \approx 36.12$, so order 10^{36}; $2019\log(365.256) \approx 5173.88$, so order 10^{5174}.

Homework

Redo practice exercise 3 with the following data: replace the initial value by 390 and the rate or common ratio by 2. Write it up in four lines: data, formula, calculation, conclusion.

Detailed solution – teacher edition

The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and give a confidence level from 1 to 4. Essential question(s) of the day: 1 and 2.

Method

1 h session: 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask quick students to produce a counterexample, a justification or a second method.

Session 55 — Reactivation: reciprocal function

Revision and consolidation

20 May 2027 • B: 1 h – C: 2 h

Session objectives

- Reactivate the reciprocal function.
- Carry out a study.
- Sketch and interpret.

Fluency – 6 to 8 minutes, no calculator

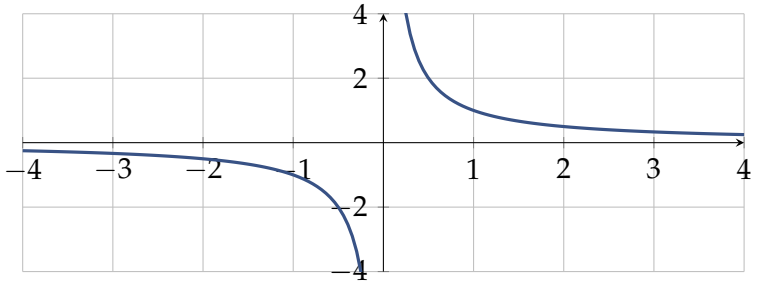
1. Write the multiplier for an increase of 10 %. $1 + \dfrac{10}{100} = 1.10$
2. State the law of total probability with two branches. $P(B) = P(A)P_A(B) + P(\overline{A})P_{\overline{A}}(B)$
3. Give the formula for the general term of an arithmetic sequence starting from u_0. $u_n = u_0 + nr$
4. Solve $10^x = 1000$. $x = 3$
5. Differentiate $x + \dfrac{4}{x}$. $1 - \dfrac{4}{x^2}$

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: reactivate the reciprocal function and carry out a study.

Investigation

Study $f(x) = x + \dfrac{16}{x}$ on $]0, +\infty[$ and interpret its minimum in an average-cost context. Produce the table of variations and a sketch.



Detailed solution – teacher edition

$f'(x) = \dfrac{(x-4)(x+4)}{x^2}$; minimum $f(4) = 8$. Have students say aloud which piece of data triggers the choice of method.

Written record

For a function containing $\dfrac{1}{x}$, start by excluding $x = 0$, then differentiate, factorise and study the sign before the variations.

Worked example

Take the data from the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer is in the teacher edition.

Detailed solution – teacher edition

$f'(x) = \dfrac{(x-4)(x+4)}{x^2}$; minimum $f(4) = 8$.

Practice – essential

1. Choose between an arithmetic sequence, a geometric sequence and an exponential function in three situations.
2. Solve an exponential equation using a logarithm.
3. Calculate a probability from a tree, then interpret it.

Detailed solution – teacher edition

Justify each choice by absolute or relative change; check positivity before taking the logarithm; multiply along a path and conclude with a sentence.

Textbook references

Further reading: Calao, p. 94, no. 1–9, then p. 96, no. 10–21. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 102, exercise 47

In Murano, a glassmaker has calculated that the production cost (in euros) of x vases is given by: $C(x) = x^2 + 30x + 400$. He wants to make as many vases as possible with the lowest possible average unit cost.

Part A

1. Calculate $C(0)$ and deduce the craftsman's fixed costs. What is the production cost of 20 vases? For 20 vases, calculate the average production cost of one vase.
2. Let $f(x)$ be the average unit cost for x vases made. For a production of at least one vase ($x \geqslant 1$), express $f(x)$ in terms of x.
3. Assume that the average unit cost can be expressed as $f(x) = x + 30 + \dfrac{400}{x}$. Calculate $f'(x)$. Show that $f'(x) = \dfrac{(x-20)(x+20)}{x^2}$. Study the sign of $f'(x)$ and deduce the variations of f. How many vases must the craftsman produce for the average unit cost to be minimal? Give this cost.

Detailed solution – teacher edition

$C(0) = 400$ €: these are the fixed costs. $C(20) = 1\,400$ €, that is 70 € per vase. For $x > 0$, $f(x) = C(x)/x = x + 30 + 400/x$. Its derivative is $1 - 400/x^2 = ((x-20)(x+20))/x^2$. On $[1;+\infty[$, it is negative up to 20, then positive. The minimum is therefore reached for 20 vases and equals 70 €.

Homework

Redo practice exercise 2 with the following data: replace the initial value by 395 and the rate or common ratio by 3. Write it up in four lines: data, formula, calculation, conclusion.

Detailed solution – teacher edition

The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and give a confidence level from 1 to 4. Essential question(s) of the day: 1, 2 and 3.

Method

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